B. Rebellion

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: constructive algorithms, greedy, two pointers | https://codeforces.com/problemset/problem/1746/B

You have an array a of size n consisting only of zeroes and ones. You can do the following operation:

- choose two indices $1 \le i , j \le n$, $i \ne j$,
- add a_i to a_j,
- remove a_i from a.

Note that elements of a can become bigger than 1 after performing some operations. Also note that n becomes 1 less after the operation.

What is the minimum number of operations needed to make a non-decreasing, i. e. that each element is not less than the previous element?

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains an integer n ($1 \le n \le 10^5$), the size of array a.

Next line contains n integers $a_1, a_2, \ldots a_n$ (a_i is 0 or 1), elements of array a.

It's guaranteed that sum of n over all test cases doesn't exceed $2 \cdot 10^5$.

Output

For each test case print a single integer, minimum number of operations needed to make a non-decreasing.

Example

Input

```
4
8
0 0 1 1 1 1 1 1
5
1 0 0 1 1
2
1 0
11
1 1 0 0 1 0 0 1 1 1 0
```

Output

```
0
1
1
3
```

Note

In the first test case, a is already non-decreasing, so you don't need to do any operations and the answer is 0.

In the second test case, you can perform an operation for $i = 1$ and $j = 5$, so a will be equal to [0, 0, 1, 2] and it becomes non-decreasing.

In the third test case, you can perform an operation for $i = 2$ and $j = 1$, so a will be equal to [1] and it becomes non-decreasing.